A Project Report on

MULTI-TASK CNN-TRANSFORMER FOR JOINT OTFS CHANNEL ESTIMATION AND DOPPLER-SENSING PREDICTION

submitted in partial fulfilment of the requirements for the course

MACHINE LEARNING
25SC2107E

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by

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DEPARTMENT OF **ELECTRONICS & COMMUNICATION ENGINEERING**

Name of Title: Multi-Task CNN-Transformer for Joint OTFS Channel Estimation and Doppler-Sensing Prediction

1. Abstract:

Orthogonal Time Frequency Space (OTFS) modulation is considered a promising waveform for future wireless communication systems, particularly in high-mobility environments such as high-speed trains, autonomous vehicles, unmanned aerial vehicles, and next-generation 5G/6G networks. Unlike conventional modulation techniques, OTFS represents wireless signals in the delay-Doppler domain, which can provide improved robustness against rapidly time-varying channels. However, accurately estimating the wireless channel and predicting Doppler-related sensing information remain challenging because high mobility produces rapid channel variations, Doppler shifts, multipath propagation, noise, and interference. Conventional channel estimation and signal-processing methods may require significant computational effort and can experience reduced performance when the channel conditions change rapidly. In addition, communication channel estimation and Doppler sensing are often treated as separate tasks, which may fail to utilize the common information available in the received OTFS signal. This project proposes a multi-task deep learning framework based on a Convolutional Neural Network (CNN) and Transformer architecture to jointly perform OTFS channel estimation and Doppler-sensing prediction. The CNN component is used to extract local spatial features and important patterns from the received OTFS signal representation in the delay-Doppler domain, while the Transformer component is designed to capture long-range relationships and dependencies among signal features. A shared feature-extraction network will learn common characteristics of the received signal, followed by separate task-specific prediction layers for channel estimation and Doppler prediction. By jointly learning both tasks, the proposed approach aims to improve feature utilization, reduce duplicate processing, and provide more reliable performance in rapidly changing wireless environments.

The proposed Multi-Task CNN-Transformer model will be evaluated using simulated OTFS communication scenarios with different signal-to-noise ratios, Doppler shifts, mobility conditions, channel delays, and multipath fading environments. The performance of channel estimation will be assessed primarily using Mean Squared Error (MSE) and Normalized Mean Squared Error (NMSE), which measure the difference between the estimated channel and the actual channel. Doppler-sensing prediction performance will be evaluated using metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and prediction accuracy or error rate, depending on whether the Doppler output is represented as a continuous value or a classified sensing state. The proposed multi-task model will also be compared with conventional estimation approaches and individual deep learning models such as standalone CNN-based or Transformer-based architectures. Additional parameters, including computational complexity, inference time, model size, convergence behavior, and robustness under varying noise and mobility conditions, will be analyzed to determine the practical effectiveness of the proposed solution. Through the combination of CNN-based local feature extraction, Transformer-based global dependency learning, and multi-task optimization, the proposed framework aims to provide accurate OTFS channel estimation and reliable Doppler-sensing prediction simultaneously, supporting efficient and intelligent wireless communication and sensing for future high-mobility communication systems.

2. SOFTWARE REQUIREMENTS:

1. **Programming Language:** Python
2. **Software:** VS Code / Google Colab
3. **Libraries:** NumPy, Pandas, Matplotlib
4. **Deep Learning:** TensorFlow / PyTorch
5. **Machine Learning:** Scikit-learn
6. **Operating System:** Windows 10/11

Signature of the Supervisor Course Instructor